

EE204 Assignment 1

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Inverting Op-Amp Circuit (Figure 2.6)

The circuit of a inverting op amp was constructed in Ltspice as seen in

I assigned appropriate values to R1,R2,and Vin to observe how they affect the output

For this section R1 was set to 10kΩ and R2 was set to 100kΩ which based on the equation below gives the gain of 10

$$A = \frac{R2}{R1} = 10$$

Equation 1

A sinusoidal signal of 0.1 amplitude and 50 Hz was applied.

As seen in the graph in Figure 2 the output was an inverted waveform that was amplified approximately 10 times confirming the gain and phase inversion of an inverting amplifier.

I also tried to recreate this circuit on a bread board unfortunately I had many issue that would not resolve I create the circuit and tried testing it but the results were not correct even when I changed components, remade the circuit ,and asked for help. This circuit is showcased in Figure 3

Inverting Op-Amp Circuit 1

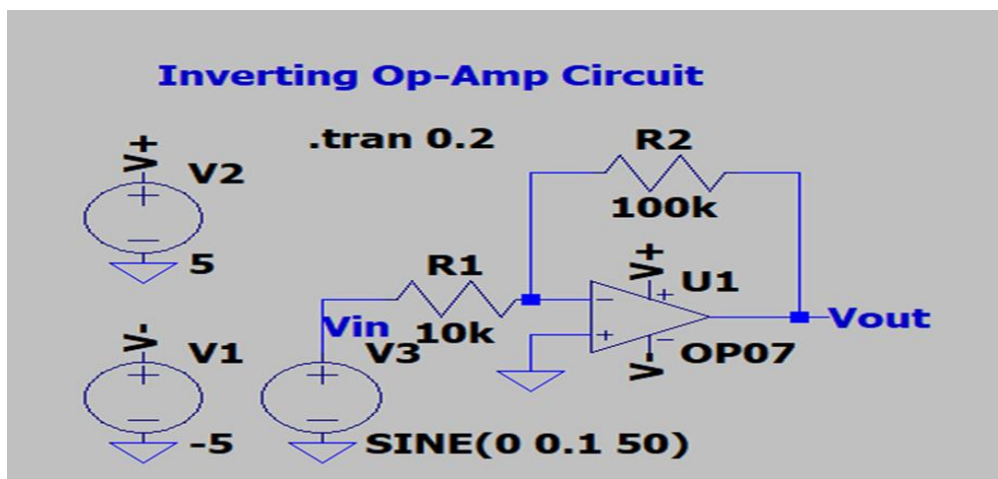


Figure 1

Vin-Vout Graph 1

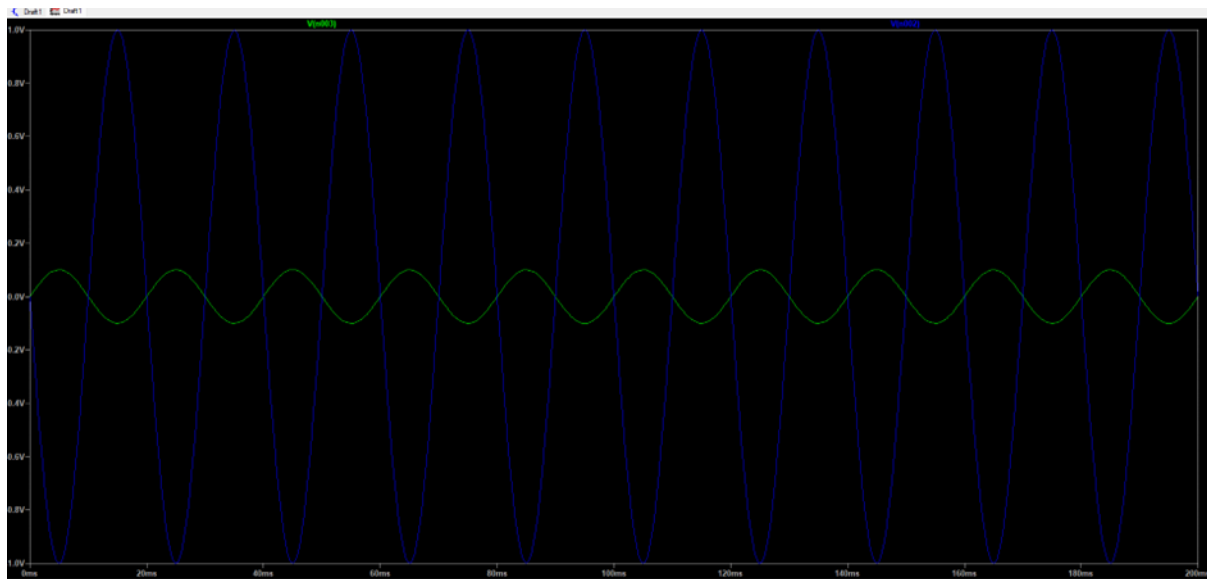


Figure 2

Circuit constructed in person 1

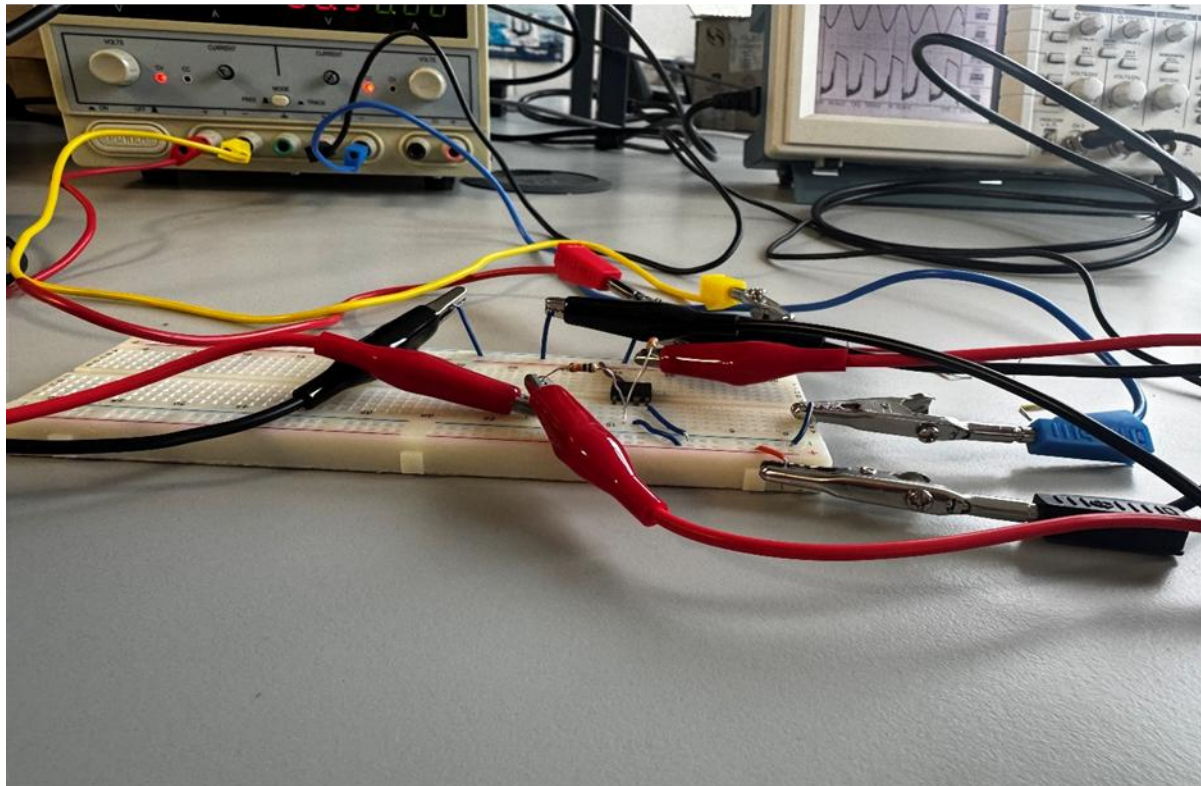


Figure 3

Modified Inverting Op-Amp Circuit (Figure 2.8)

In this modified inverting op amp circuit simulation i set the values of R1 to 10kΩ , R2 to 100kΩ R3 to 8.2k Ω and R4 to 5k Ω. with a 0.1v sine wave input with a 50Hz frequency as showcased in Figure 4. The theoretical gain of this circuit is given by equation 2 below.

$$\frac{V_o}{V_i} = -\frac{R_2}{R_1} \left(1 + \frac{R_4}{R_2} + \frac{R_4}{R_3} \right) = \frac{100K}{10K} \left(1 + \frac{5K}{100K} + \frac{5K}{8.2K} \right) = -10(1 + 0.05 + 0.6098) = -16.6$$

Equation 2

$$V_o = 0.1 \times 16.6 = 1.6V$$

Equation 3

The results verified the effects of the modified inverting op amp. The amplified output and phase correspond to the theory given in chapter 2 where the gain of Vo was approximately 1.6V as showcased in Figure 5.

the modified inverting amplifier improves on the basic inverting amplifier by adding 2 resistors which are used to balance input bias currents and minimise DC offset at the output therefore making the circuit more accurate and stable.

Modified Inverted Op-Amp Circuit 1

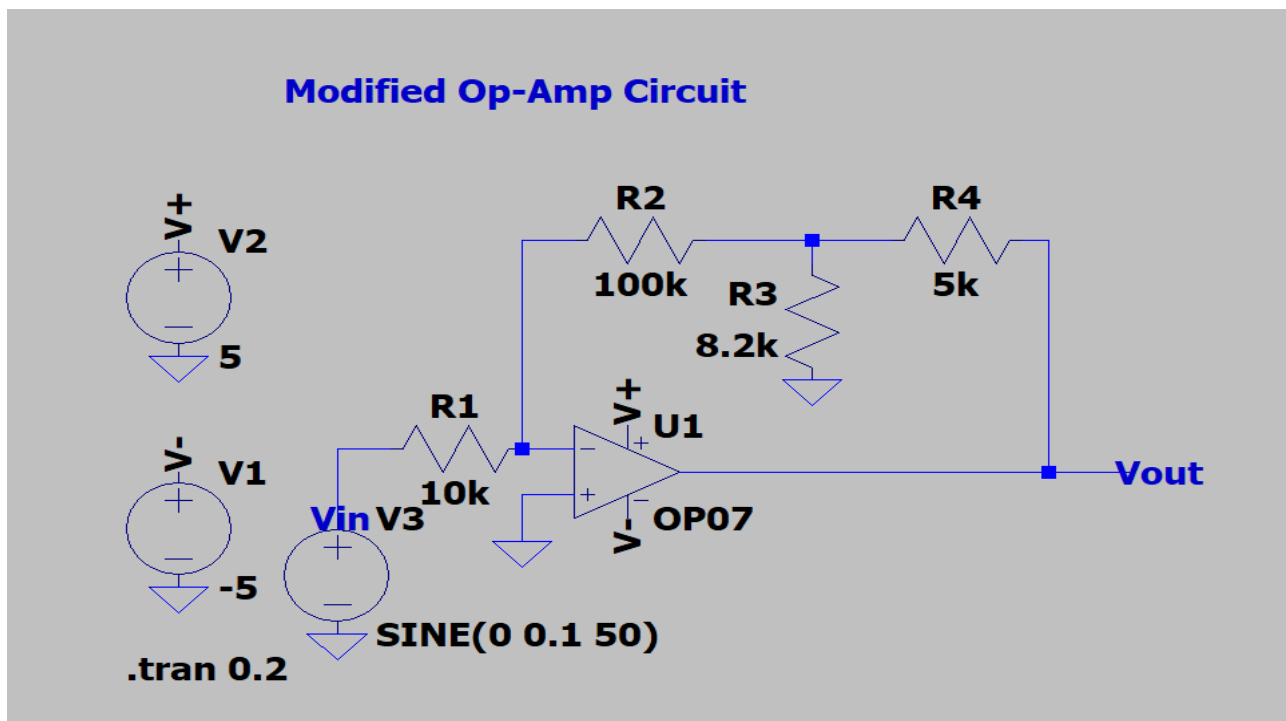


Figure 4

Vin - Vx - Vout Graph 1

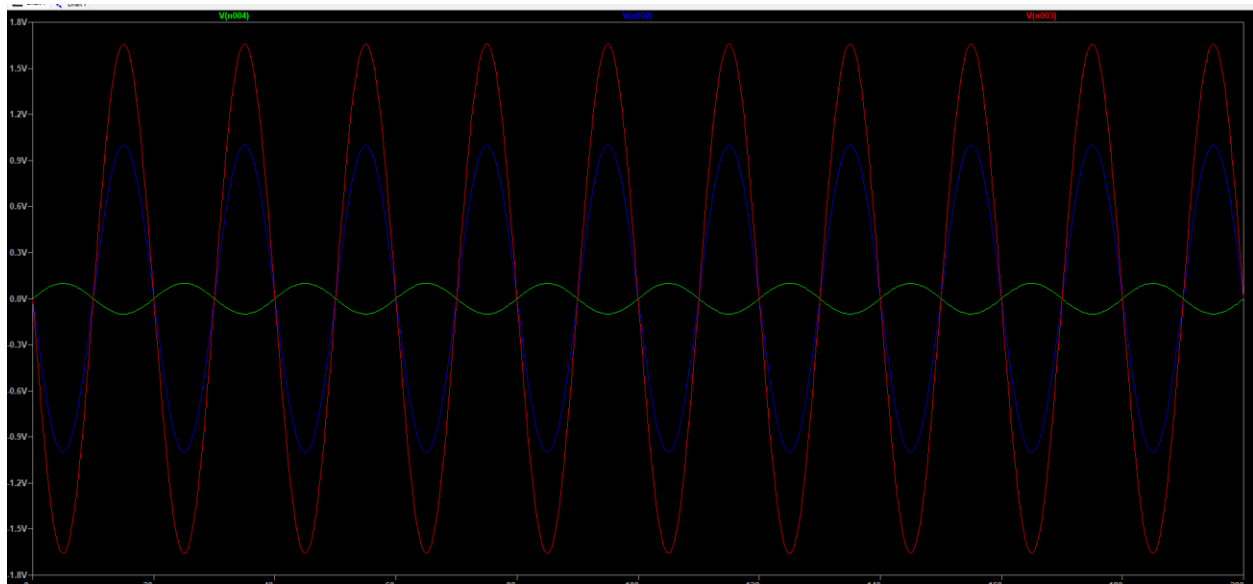


Figure 5

Weighted Summer Op Amp Circuit (Figure 2.10)

In this weighted summer op-amp circuit simulation, I set the resistor values to $R_1 = 10k\Omega$, $R_2 = 20k\Omega$, $R_3 = 40k\Omega$, and the feedback resistor $R_f = 10k\Omega$.

Three input sine wave sources were used, each with an amplitude of 0.1 V but at different frequencies of 50 Hz, 100 Hz, and 200 Hz respectively, as showcased in Figure 6

The theoretical output voltage of the weighted summer is given by Equation 4 below:

$$V_o = -R_4 \left(\frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3} \right)$$

Equation 4

With the values substituted the measured output amplitude was approximately 0.35V, confirming the accuracy of the gain and demonstrating a proper weighted summer of the three inputs as showcased in Figure 7 and **Error! Reference source not found..**

Weighted Summer Circuit 1

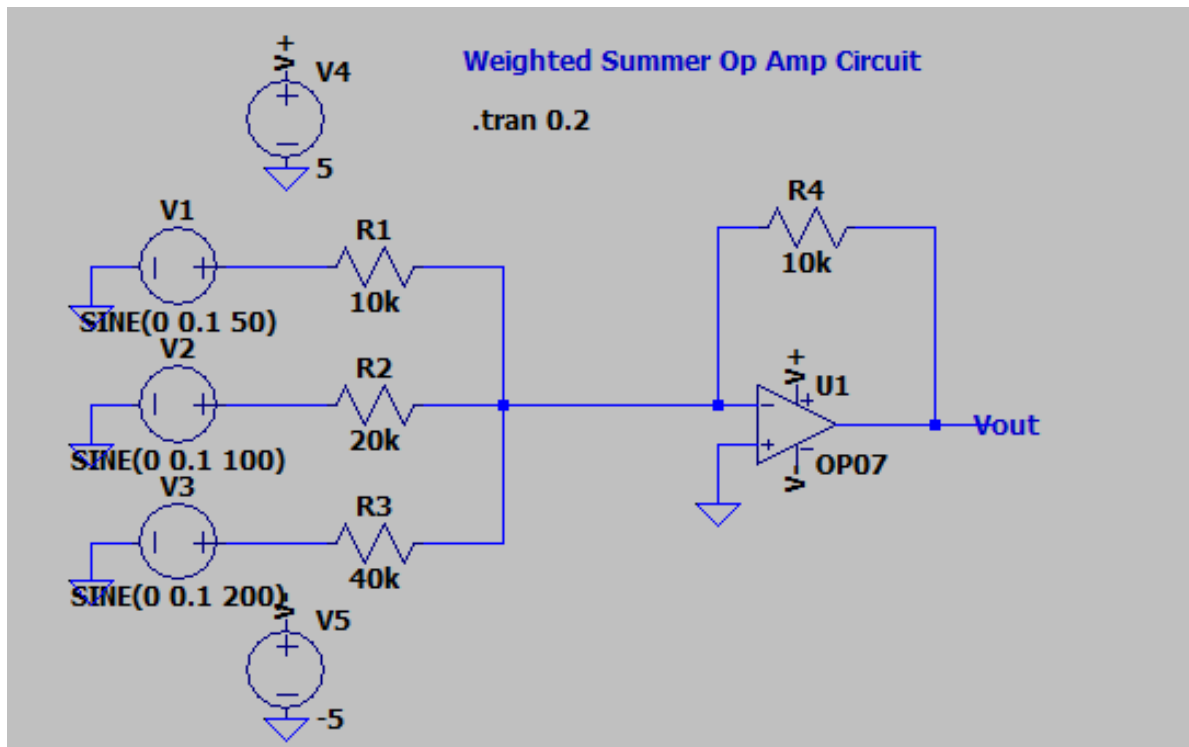


Figure 6

VR1 - VR2 AND VR3 - VRF 1

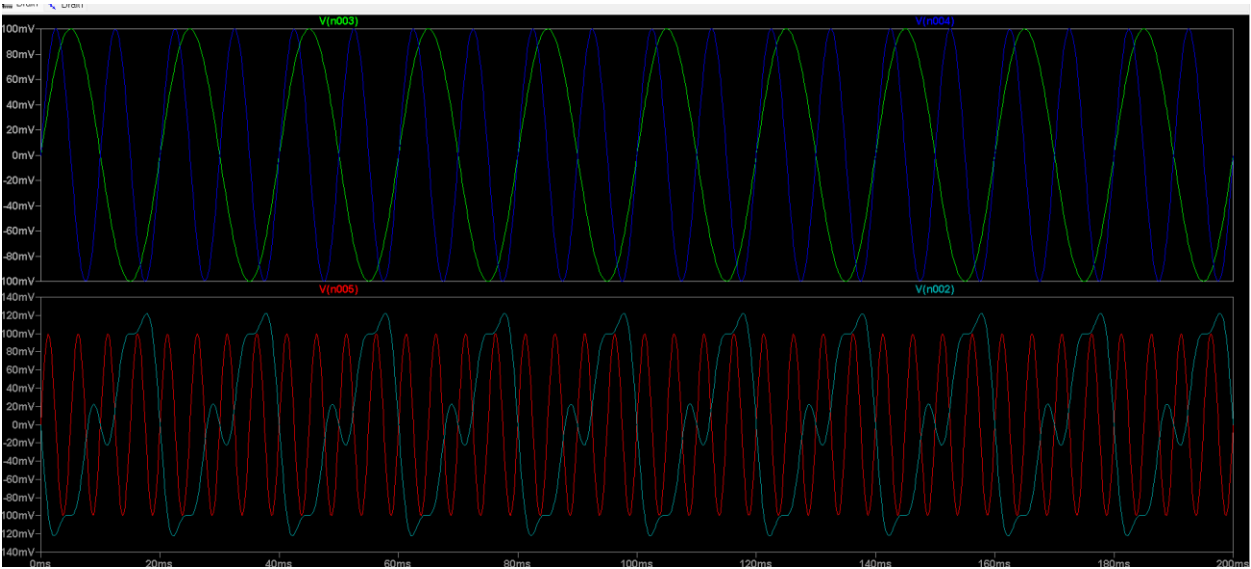


Figure 7

Non-Inverting Op Amp Circuit (Figure 2.13)

In this non-inverting op-amp circuit simulation, I set the resistor values to $R_1 = 10k\Omega$ and $R_2 = 90k\Omega$. A 0.1 V sine-wave input at 50 Hz was applied as showcased in Figure 8.

The gain obtain was through Equation 5 below:

$$\frac{V_o}{V_i} = 1 + \frac{R_2}{R_1}$$

Equation 5

The simulated waveform verified the theoretical behavior of the non-inverting amplifier. The output was a clean sine wave, in phase with the input and amplified by a factor of 10, matching the expected theoretical gain. The measured output amplitude was approximately 1 V, confirming that $A_v = 10$, as shown in Figure 9

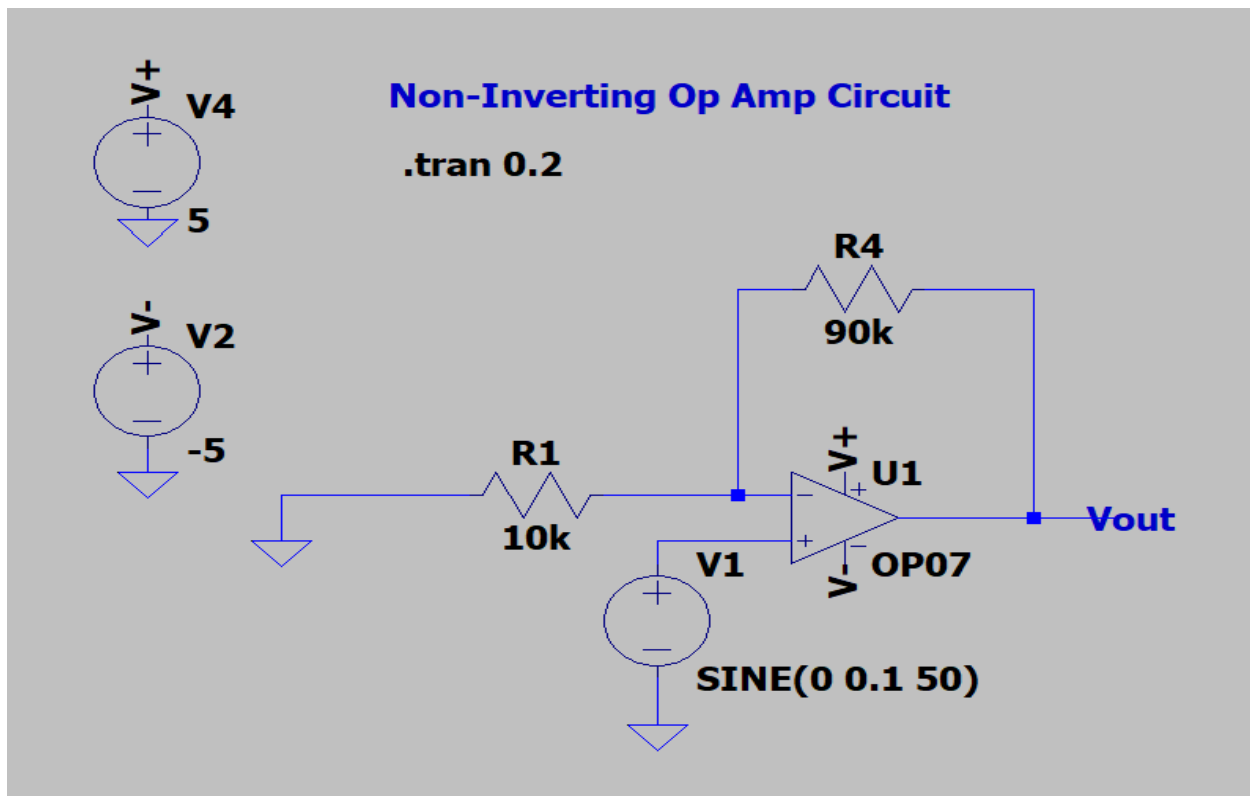


Figure 8

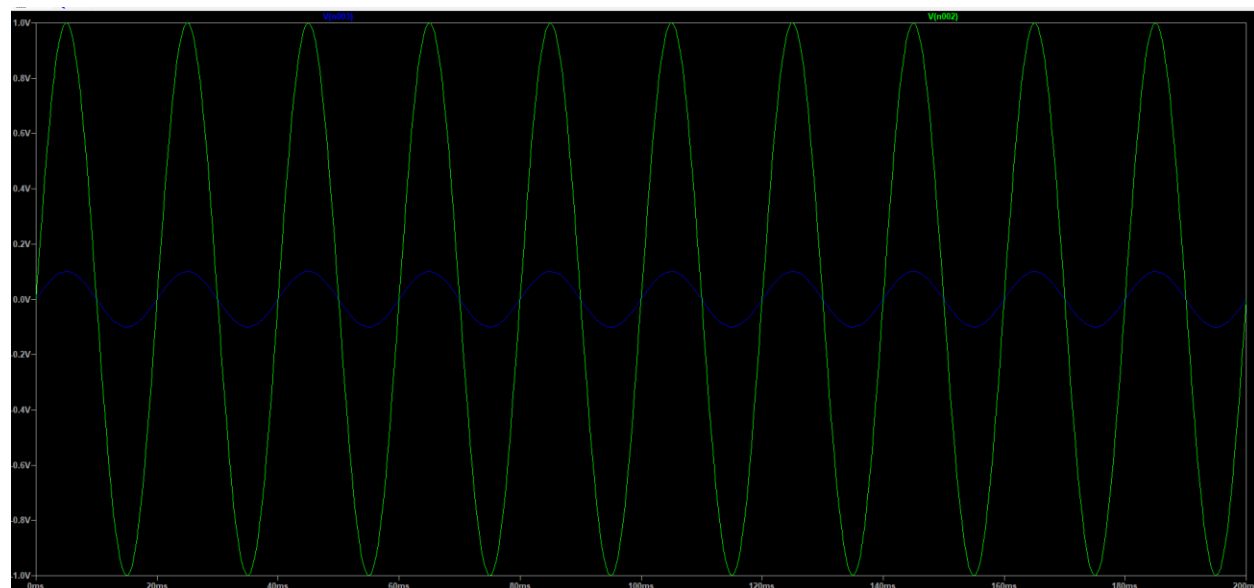


Figure 9

Voltage Follower Circuit (Figure 2.14)

In this voltage-follower circuit simulation, the output was directly connected to the inverting input of the op-amp, forming a unity-gain configuration.

The same 0.1 V, 50 Hz sine-wave input signal was applied to the non-inverting input terminal as in Figure 2.13, as showcased in Figure 10`

The gain for the voltage gain follow is given by Equation 5 and since there is no resistors and R1 theoretically became infinity the the gain was therefore 1.

This means the output voltage is identical to the input in both amplitude and phase the output voltage perfectly overlapped the input with no phase shift or amplification. Both signals were 0.1V, verifying the unity gain and confirming that the circuit acts as a buffer with very high input impedance and very low output impedance as showcased in Figure 11

Voltage Follower Circuit 1

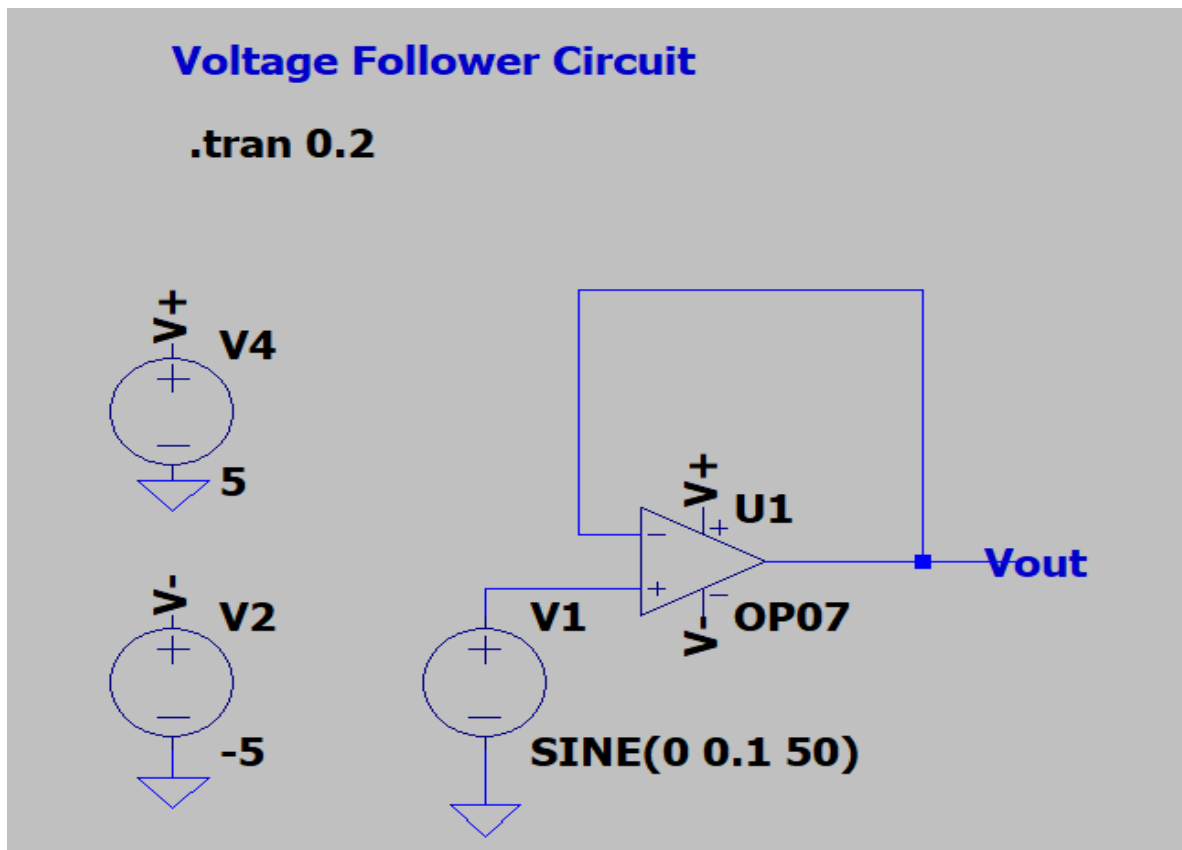


Figure 10

Vin-Top Vout - Bottom 1

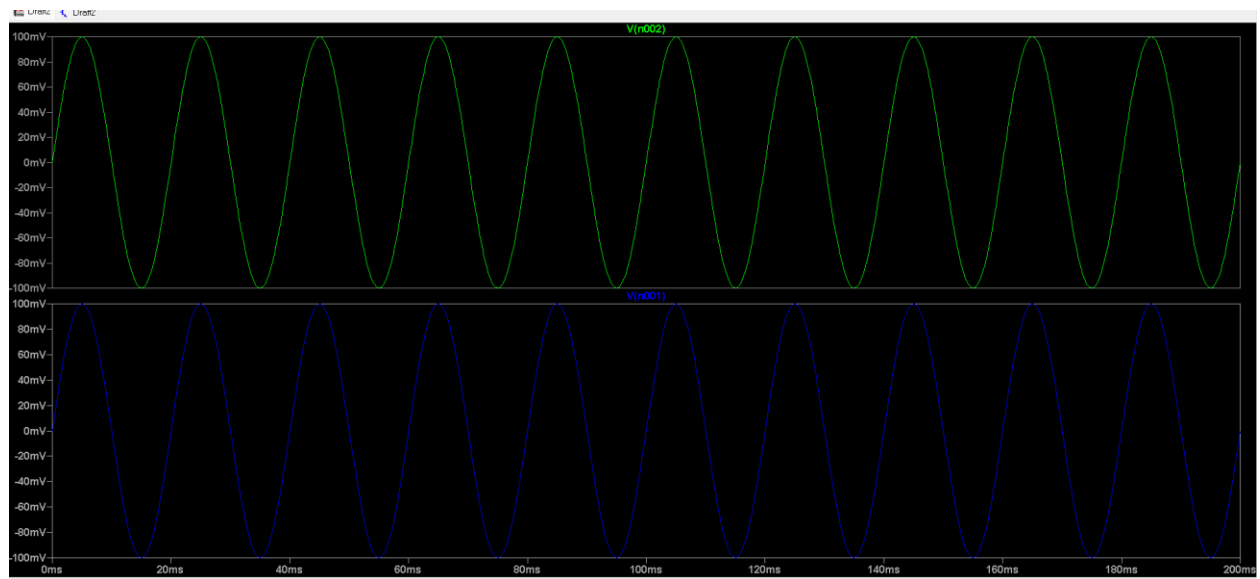


Figure 11